Experiment No. 2

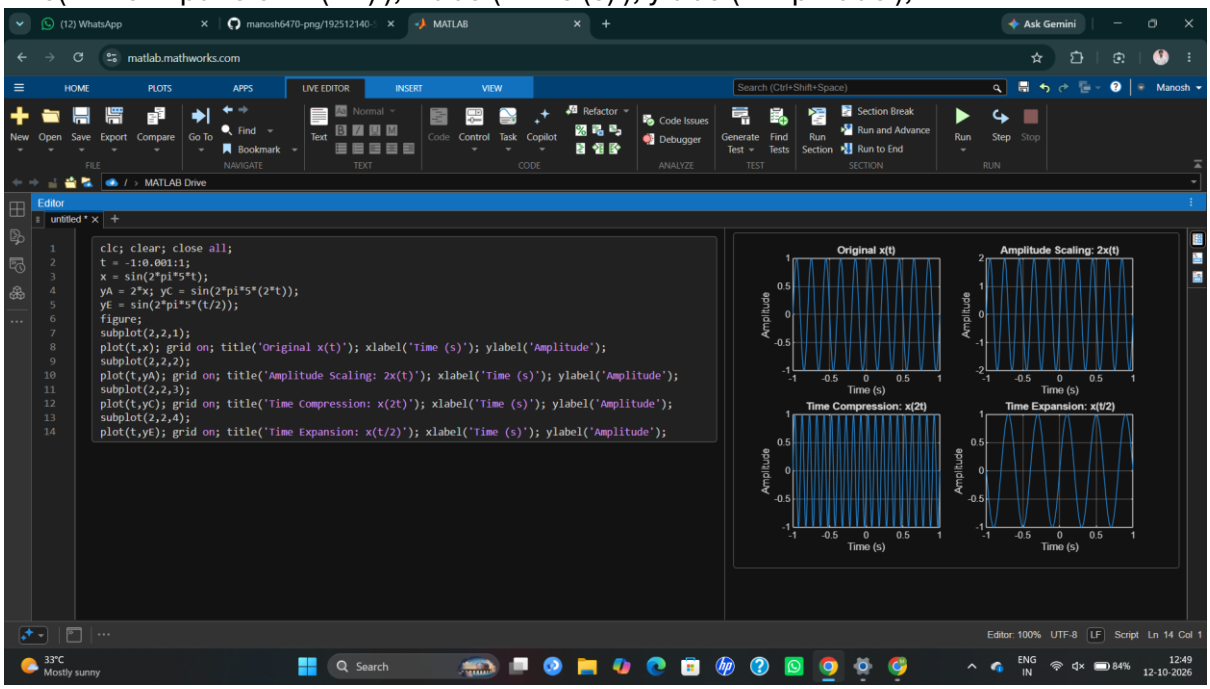
Signal Operations and Transformations

(i) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

MATLAB Coding

% Experiment 2(i): Amplitude and Time Scaling

```
clc; clear; close all;
t = -1:0.001:1;
x = sin(2*pi*5*t);
yA = 2*x;
yC = sin(2*pi*5*(2*t));
yE = sin(2*pi*5*(t/2));
figure; subplot(2,2,1);
plot(t,x); grid on;
title('Original x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,2,2); plot(t,yA);
grid on;
title('Amplitude Scaling: 2x(t)');
xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,3); plot(t,yC); grid on;
title('Time Compression: x(2t)');
xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,4); plot(t,yE); grid on;
title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude');
```



(iii) To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics

MATLAB Coding

% Experiment 2(iii): Addition and Multiplication of Two Sequences

```
clc; clear; close all;
n = 0:2; x = [1 2 1]; y = [1 1 2];
zSum = x + y;
zProd = x .* y; figure;
subplot(2,2,1); stem(n,x,'filled'); grid on; title('x[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,2); stem(n,y,'filled'); grid on; title('y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,3); stem(n,zSum,'filled'); grid on; title('Addition x[n]+y[n]'); xlabel('n');
ylabel('Amplitude');
subplot(2,2,4); stem(n,zProd,'filled'); grid on; title('Multiplication x[n]y[n]'); xlabel('n'); y
label('Amplitude');
```

